

AMPGen — AMP Challenge Submission Report

A two-stage generative pipeline for antimicrobial peptide design

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Backbone: ESM2-35M (facebook/esm2_t12_35M_UR50D) · Compute: Modal, single H100

This report documents the AMPGen *submission*: the deliverables, the data behind them, the generator and predictor that produce and rank the library, and how our internal metrics map to the challenge’s scoring. It is the submission-facing companion to the AMPGen method paper (reports/AMPGen_Report.pdf), which carries the full training methodology and ablations.

1 Submission summary

AMPGen is a two-stage pipeline. **Stage-1** is an activity-blind generator (a domain-adapted masked-language model over the full AMP corpus, used as a Gibbs mutation proposer) that produces natural-like candidate peptides. **Stage-2** is a dual-endpoint predictor on the same frozen encoder that scores each candidate for antibacterial activity (per-species MIC) and safety (human-RBC HC50), and ranks the library. Deliverables:

- A **50,000-sequence FASTA library** of designed peptides satisfying the challenge constraints (20 standard amino acids, length 8–50, linear, free termini).
- A **ranked top-100 list**, ordered by the predictor’s activity/selectivity score, with every entry $\leq 80\%$ identical to any DBAASP/dbAMP/APD reference (novelty gate).
- The generator and predictor weights, training data manifest, and this report.

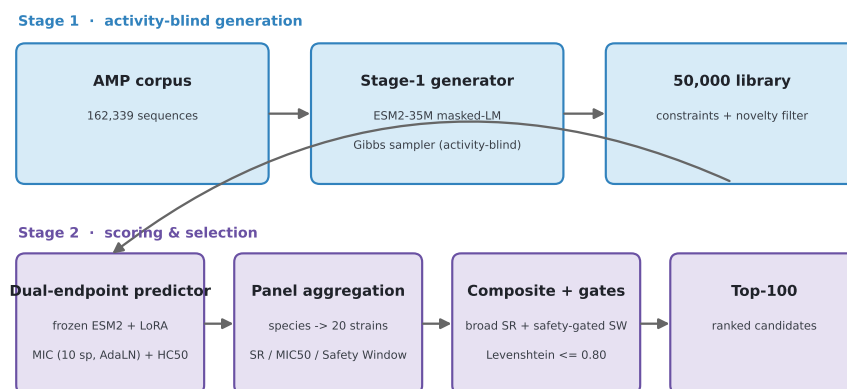


Figure 1: AMPGen submission pipeline. Stage 1 (activity-blind generation): the domain-adapted ESM2-35M masked-LM Gibbs sampler turns the AMP corpus into a constraint- and novelty-filtered 50,000-sequence library. Stage 2 (scoring & selection): the dual-endpoint predictor (frozen ESM2 + LoRA; 10-species AdaLN MIC head + HC50 head) scores each candidate; panel aggregation maps species predictions to the 20-strain challenge panel (Success Rate / MIC50 / Safety Window); a safety-gated composite plus the strict Levenshtein novelty gate yields the ranked top-100. The whole pipeline is one reproducible command, `uv run generate`.

2 Challenge task and scoring metrics

Submissions are scored by the organisers’ **seqme** library against a fixed **20-strain panel** (15 Gram-negative, 5 Gram-positive; an 8-strain MDR ESKAPE subset) across **five categories**. The table

gives what each category rewards and the underlying quantities.

Table 1: Challenge scoring categories and the quantities they reward. Potency threshold = 16 μ M; MIC ceiling = 64 μ M; HC50 ceiling = 128 μ M.

Category	What it rewards	Underlying quantity
Broad-Spectrum	Activity across the whole 20-strain panel	Success Rate over all strains
Gram-Positive	Activity on the 5 Gram+ strains	Success Rate / MIC50 (Gram+)
Gram-Negative	Activity on the 15 Gram− strains	Success Rate / MIC50 (Gram−)
MDR ESKAPE	Activity on the 8-strain drug-resistant subset	Success Rate (MDR subset)
Optimal Selectivity	Potent <i>and</i> non-toxic to human cells	Safety Window = HC50 / MIC50

Function of each scoring quantity.

- **Success Rate** — fraction of panel strains for which a peptide reaches MIC $\leq 16 \mu$ M (the potency threshold). The core “does it work” measure; drives the four activity categories.
- **MIC50 / MIC90** — median / 90th-percentile MIC across the relevant strain set; summarise potency depth and breadth.
- **Safety Window (HC50/MIC50)** — how much more concentrated a peptide must be to lyse human red blood cells than to kill bacteria; the therapeutic-index proxy that defines the Optimal-Selectivity category.

3 Data sources and scale

All three data assets are unions of curated public databases, de-duplicated and split leak-free (CD-HIT at 80% identity, whole clusters assigned to train or test so no test sequence is $>80\%$ identical to any training sequence).

Table 2: Data assets behind the submission. MIC and HC50 feed the two predictor endpoints; the generator corpus is the Stage-1 pretraining set.

Asset	Sources (union)	Seqs	Meas.	Notes
Generator corpus	AMPSphere (physchem-matched) + DRAMP + DBAASP + dbAMP + APD + 7,059 activity-validated peptides	162,339	—	all-AMP MLM pretrain
MIC (10 species)	GRAMPA + ampbench-v0.7 master + AMPBench-MT	13,123	40,874	$\log_{10} \mu$ M; 14 flags free-termini
HC50 (human RBC)	ampbench master hc50_human (DBAASP+HemoPI2+Hemolytik+QMAP+DRAMP) + Hemolytik2 + DRAMP text	4,402	4,402	14.2% right-censored

4 Stage-1 — Generator

The generator is a domain-adaptive masked-LM pretrain of ESM2-35M on the 162,339-sequence all-AMP corpus (variable-rate span/scattered masking, $1/t$ -reweighted cross-entropy, 3 epochs), used as a temperature-sampled mutation proposer. It is judged on whether its output is *natural-like* on the axes that matter for AMPs (physicochemical marginals and an independent language-model naturalness score), not on exact-residue recovery.

Table 3: MIC measurement scale per panel species (post-union, 10 species). AMPBench-MT contributes the DBAASP rare-species MIC our earlier GRAMPA+master build did not include, multiplying the low-data species 3–13 $\times$.

Species	Gram	measurements	panel strains
<i>E. coli</i>	–	11,143	5
<i>S. aureus</i>	+	9,776	2
<i>P. aeruginosa</i>	–	6,977	3
<i>B. subtilis</i>	+	3,139	1
<i>S. enterica</i>	–	2,393	2
<i>K. pneumoniae</i>	–	2,353	2
<i>A. baumannii</i>	–	2,025	2
<i>E. faecalis</i>	+	1,779	1
<i>E. faecium</i>	+	771	1
<i>E. cloacae</i>	–	518	1
Total		40,874	20

Table 4: Generator (Stage-1) held-out metrics. Reference = held-out natural AMPs; pseudo-perplexity is from an independent ESM2-150M, distinct from the generator backbone.

Metric	init	final	reference / comparator
Validation loss	2.684	2.194	—
Exact masked-token recovery	0.213	0.297	chance 0.05 ($> 3\times$)
Novelty, fraction $\leq 80\%$ identity	—	100%	challenge novelty gate
Embedding diversity (mean pairwise cos-dist)	—	0.0977	natural 0.1009 (matched)
Pseudo-perplexity, median	—	17.12	natural 17.01 (near-identical)
Wasserstein: len / charge / GRAVY / μ H / hydrophobic	—	0.000 / 0.099 / 0.014 / 0.005 / 0.002	(lower = closer)

The generator reproduces the natural AMP marginal (physchem Wasserstein near zero on every axis, pseudo-perplexity matched to natural) and passes the novelty gate at 100%, i.e. it neither collapses to a degenerate mode nor drifts out-of-distribution.

5 Stage-2 — Dual-endpoint predictor

On the frozen Stage-1 encoder we train LoRA ($r=8$, $\alpha=16$, last 4 attention layers; 0.79 M trainable params, 2.3%) feeding two heads: a **species-conditioned MIC head** (AdaLN over 10 species) predicting $-\log_{10}\text{MIC}$, and a **single HC50 head** predicting $\log_{10}\text{HC50}$. Both are trained to *absolute* $\log_{10} \mu\text{M}$ with a composite loss (pairwise ranking + Huber magnitude regression); the HC50 loss is censoring-aware (right-censored “safe-tail” measurements penalised one-sidedly). The two heads are trained as **separate** models (shared frozen encoder, own LoRA) — multitask training gave no MIC gain and a clear HC50 loss (the small-task seesaw), so separate is the final configuration.

5.1 Version progression (v1 $\rightarrow$ v4)

[†] **The apparent v3 dip in Table 5 is a split artifact, not a regression;** v3 and v4 are both on this harder re-clustered split. The v1 $\rightarrow$ v3 rows are each on their *own* CD-HIT split; adding 16,821 MIC rows re-clustered everything, so v3’s honest test set differs from v2’s and is harder. Proof: HC50’s data is *unchanged* v2 $\rightarrow$ v3 yet its honest ρ moved $0.503 \rightarrow 0.421$ (-0.082) purely from the re-split. To measure the data effect cleanly we ran a **controlled ablation on one fixed split** (Table 7): same held-out test, training data $\pm$ AMPBench-MT. There the merge **improves overall**

Table 5: Predictor performance across iterations, overall MIC and HC50 Spearman ρ and Pearson r on *both* splits (homology = honest, random = leaky reference). v1 = dual-endpoint baseline; v2 adds rare-species sampling rebalance and censoring-aware HC50 loss; v3 adds the AMPBench-MT data merge; v4 adds conservative MIC de-noising (MIC head) and a listwise soft-Spearman ranking term (HC50 head). v3 and v4 share the same re-clustered honest test (so v4–v3 is the attributable gain); v4 lifts honest MIC +0.026 and HC50 +0.060 over v3.

Ver	Split	MIC ρ	MIC r	HC50 ρ	HC50 r
v1	homology	0.578	0.592	0.456	0.478
v1	random	0.645	0.645	0.597	0.624
v2	homology	0.585	0.597	0.503	0.522
v2	random	0.637	0.627	0.563	0.577
v3	homology	0.494 [†]	0.512 [†]	0.421 [†]	0.459 [†]
v3	random	0.619	0.639	0.592	0.604
v4	homology	0.520[†]	0.536[†]	0.481[†]	0.492[†]
v4	random	0.624	0.646	0.603	0.596

MIC ρ **0.450**→**0.494** (+0.044) and the rare species by +0.140 (strain-weighted) — *A. baumannii* +0.30, *E. faecium* +0.25, *K. pneumoniae* +0.11 — with the big-3 flat (+0.004). So the AMPBench-MT data **genuinely helps**, most where the challenge scores it (MDR/Gram), and the deployed model is the v2 configuration trained on the full merged corpus.

5.2 Per-species MIC (both splits)

Table 6: Per-species MIC Spearman ρ and Pearson r for the v2-configuration model on *both* splits (homology = honest, random = leaky reference). The low- n Gram+ species (*E. faecalis*, *E. cloacae*, *E. faecium*) carry wide confidence intervals and should be read as noisy estimates, not point values.

Species	homology (honest)		random	
	ρ	r	ρ	r
<i>E. coli</i>	0.598	0.605	0.653	0.647
<i>S. aureus</i>	0.550	0.551	0.606	0.606
<i>P. aeruginosa</i>	0.631	0.638	0.655	0.658
<i>B. subtilis</i>	0.546	0.571	0.591	0.569
<i>S. enterica</i>	0.578	0.722	0.675	0.699
<i>K. pneumoniae</i>	0.590	0.751	0.655	0.686
<i>E. faecalis</i>	0.255	0.411	0.504	0.444
<i>E. cloacae</i>	0.356	0.514	0.463	0.565
<i>E. faecium</i>	0.469	0.516	0.460	0.456
<i>A. baumannii</i>	0.681	0.651	0.612	0.647

6 Our evaluation metrics and their function

The predictor is scored with four complementary metrics, on two splits, against a mean-predictor baseline — so that a number can never be read as good without its honest reference.

- **Spearman ρ** (ranking, calibration-free) — how well the score *orders* peptides by activity/safety. This is what the **top-100 ranking** deliverable depends on.

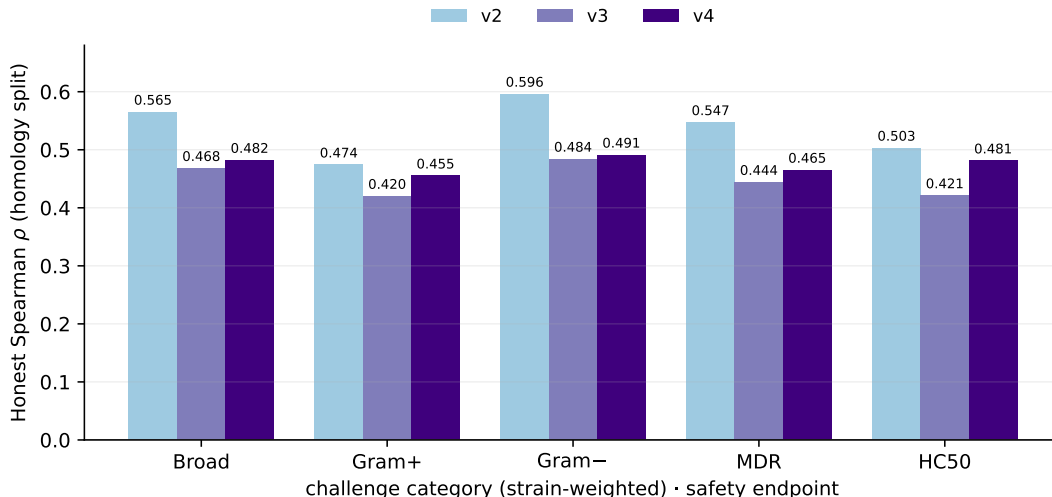


Figure 2: Version progression v2 → v3 → v4, category-level *honest* ranking. Bars are category-level Spearman ρ on the honest (homology) split — strain-weighted per-species MIC via the panel aggregation of §7, plus the HC50 safety endpoint — coloured by version. Random-split numbers are omitted here (they are in Table 5). v2 is evaluated on its own pre-re-clustering split; v3 and v4 share the common, harder re-clustered honest test, so **v4 vs v3 is the apples-to-apples comparison**: v4 (data-clean MIC head + soft-rank HC50 head) improves *every* category over v3, most on HC50 (+0.060) and the resistance-relevant Gram+/MDR. v2’s higher bars reflect its easier split, not a regression at v3 (see the † note).

- **Pearson r** (magnitude, on absolute $\log_{10}\mu\text{M}$) — whether predicted values track true values in *absolute* terms. Required because the challenge scores an absolute $16\mu\text{M}$ potency threshold and the HC50/MIC50 safety window; a rank-only model cannot supply these. We measure r on *raw* values, not by inverting a ranking score. Because $\rho \approx r$ (not $\rho \approx 2r$), the model genuinely calibrates magnitude.
- **RMSE / MAE** ($\log_{10}\mu\text{M}$) — absolute error, reported against the mean-predictor’s RMSE (its ceiling when $\rho=r=0$).
- **Split.** The *homology* split (CD-HIT-80% whole-cluster) is the honest, deployment-relevant number; the *random* by-sequence split leaks homologs across train/test and is the inflated, literature-comparable number. We always report both.

7 Panel aggregation — species to category scores

The predictor emits one MIC per species and one HC50; the challenge scores a 20-strain panel. The aggregation (`metrics/aggregate.py`) broadcasts each species’ predicted MIC to the panel strains it represents (PANEL_STRAIN_COUNTS: *E. coli*×5, *P. aeruginosa*×3, ...) and computes, per category, **Success Rate** (fraction of strains with MIC $\leq 16\mu\text{M}$), **MIC50/MIC90** (strain-weighted percentiles), and, for Optimal-Selectivity, **Safety Window** = HC50 / MIC50 — all with the challenge ceilings (MIC $64\mu\text{M}$, HC50 $128\mu\text{M}$) applied. Categories: Broad (all 20 strains), Gram+ (5), Gram− (15), MDR (the 8-strain ESKAPE subset). The same category weighting yields the category-level ranking ρ shown per version (v2–v4, honest split) in Figure 2 and used for the fair data ablation in Table 7.

8 Decision and status

Final predictor (v4): the separate-mode configuration (species-rebalanced sampling + censoring-aware HC50 loss) trained on the **full merged corpus incl. AMPBench-MT**, with two v4 levers

Table 7: Fair data ablation, category-level Spearman ρ on the *identical* fixed v3 split (same test; only training data differs). Adding AMPBench-MT lifts every MIC category, most on the resistance-relevant MDR and Gram+; HC50 is unchanged (its data is identical in both arms).

Training data	Broad	Gram+	Gram−	MDR	HC50
Arm A — v2 data (GRAMPA+master)	0.394	0.323	0.418	0.345	0.428
Arm B — v3 data (+AMPBench-MT)	0.468	0.420	0.484	0.444	0.421
Δ (B−A)	+0.074	+0.097	+0.066	+0.099	−0.007

chosen *per endpoint* from a controlled 5-arm ablation on one fixed split: the MIC head uses **conservative de-noising** (drop cross-source contradictions > 3 doubling dilutions), and the HC50 head adds a **listwise soft-Spearman ranking** term. Both were selected because stacking all levers interfered, whereas each endpoint’s best single lever is a clean, attributable win. **Honest (homology-split) metrics:** MIC ρ **0.520** / r **0.536**, HC50 ρ **0.481** / r **0.492** — +0.026 MIC and +0.060 HC50 over the v3 baseline on the identical test. Supporting findings: (i) the censoring-aware HC50 loss and the soft-rank term are both clear HC50 wins; (ii) the controlled data ablation (Table 7) shows AMPBench-MT improves MIC ranking (overall +0.044, rare species +0.140, concentrated in the MDR/Gram categories the challenge weights most); (iii) a physicochemical-feature branch was diagnosed out (incremental r +0.015 MIC / ~ 0 HC50 — ESM2 already subsumes the descriptors). A dataset-specific noise ceiling ($r_{\max} = \sqrt{1 - \sigma_e^2 / \text{Var}(y)}$, from cross-source measurement collisions) places these at MIC ~ 0.79 within-cluster / 0.92 assay-oracle and HC50 ~ 0.84 / 0.80–0.90, showing HC50 retains the larger headroom.

Submission status: the fixed-seed (42) 50,000-sequence library and the novelty-gated top-100 (ranked by the aggregated selectivity score of §7) have been **generated and pass all local challenge checks** — 50,000 valid unique sequences, no exact match to the reference set, and every top-100 entry at Levenshtein ratio ≤ 0.80 to any reference. Remaining: upload to the competition and a **seqme** scoring mirror for local validation.